

WE WILL FINALIZE THE TITLE AFTER THE REVIEW

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ABSTRACT

Large Language Models (LLMs) are increasingly being used in the education sector to support educators and domain experts in generating and classifying Course Learning Outcomes (CLOs) according to Bloom's Taxonomy. However, manual alignment of learning objectives with Bloom's cognitive levels remains highly subjective, time-consuming, and labor-intensive, and is often driven by superficial verb matching rather than genuine cognitive depth. These challenges have motivated researchers and educators to explore automated approaches for Bloom's Taxonomy classification. Earlier studies investigated traditional machine-learning and deep-learning models, while more recent research has shifted toward LLM-based approaches. Nevertheless, existing studies remain fragmented, characterized by heterogeneous datasets, inconsistent evaluation metrics, and varying interpretations of Bloom's cognitive levels, which hinder meaningful cross-study comparability. To address these challenges, we conducted a Systematic Literature Review (SLR) following PRISMA guidelines, synthesizing evidence from 76 empirical studies published between 2018 and 2025. The review examines the effectiveness of LLMs across Bloom's cognitive levels, the use of commonly used datasets, comparative performance across model architectures, and key limitations and future research opportunities. Our findings provide a consolidated view of current practices and identify critical gaps that must be addressed to enable robust and scalable Bloom-aligned educational design.

INDEX TERMS Keywords: Bloom's Taxonomy, Large Language Models (LLMs), Artificial Intelligence, AI, AI in Education, Learning Objectives: Classification, Educational Assessment, Transformer Models Prompt Engineering, Systematic Literature Review.

I. INTRODUCTION

A. CONTEXT AND BACKGROUND

The systematic implementation of Artificial Intelligence (AI) and Machine Learning (ML) into modern educational systems has radically changed the organization and execution of the teaching, learning, and assessment processes. The early educational technologies were mainly based on the idea of digitization, and they were represented by learning management systems, online vectors, and automated grading systems. However, the recent introduction of LLMs is a qualitative change, models can now solve tasks such as content generation, explain, provide feedback, and design assessments, among others. Therefore, the idea of LLMs is increasingly being viewed not as tools of instruction only but rather as participants of pedagogical processes.

However, the pedagogical value of LLMs cannot be evaluated using only the concepts of solely by completion. In contrast to the traditional software systems, LLMs are inherently probabilistic in nature, producing results based on the patterns which they have learned, and not according to strict rules. In this regard, it is of utmost importance that the effectiveness of LLMs in the education setting hinges on how well their outputs align with better-grounded pedagogical frameworks that define them adopted standard for structuring educational goals. Bloom's Taxonomy, in the 1950s and later revisited by Anderson and Krathwohl, is a hierarchy of cognitive processes arranged in a cumulative hierarchy that forms the basis of educational design, assessment design, and curriculum alignment in the fields and educational levels.^{n2001taxonomy.re} commonly referred to as Lower-

Order Thinking Skills (LOTS) and Higher-Order Thinking Skills (HOTS) [1]:

- **Foundational Levels (LOTS):** It begins with the level of 'Remembering', where the factual knowledge is recalled, and 'Understanding', where meaning is formed out of instructional messages. It is then followed by 'Applying' where learners apply acquired knowledge or procedures in both familiar and new situations. [2].
- **Advanced Levels (HOTS):** The higher levels involve more complex forms of cognition: an action like 'Analyzing' presupposes dissecting something to find relationships and hidden patterns; an action like 'Evaluating' presupposes making a judgment on the basis of explicit criteria and standards; and an action like 'Creating' is the most advanced form of cognitive activity, and it requires learners to assemble something new out of pieces. [3], [4].

Over the decades this hierarchical structure has guided teachers in their development of learning beyond mere surface memorization and on into higher order analytical, evaluative and creative thought. Bloom's Taxonomy is used as the pedagogical standard according to which the learning goals, evaluation tools, and the mastery of the students are systematically evaluated as illustrated in Figure-1

In the modern environment of Generative Artificial Intelligence (GenAI), the Bloom's Taxonomy has shifted its application as a guide used in the manual instruction process to a computational standard of assessing the AI-based educational processes. [5]. The LLMs, such as GPT-4 and Gemini, are theoretically applicable to the automation of adaptive learning pathways through the dynamically generated instructional content adjusted to the specific cognitive level of the existing taxonomy. [6]. etical abilities of LLMs do not manifest themselves in practice equally well. Recent findings indicate that there is indeed a cognitive alignment gap: despite the fact that LLMs demonstrate good performance in challenges oriented to (LOTS) (such as recall-based quizzes and simple explanation tasks), and they are highly unstable and inconsistent in terms of performance in HOTS, especially in the ability to differentiate closely related levels like 'Analyzing' 'Creating' and 'Evaluating' [7], [8].

Such a mismatch raises substantiated pedagogical concerns. Unless the AI systems are applied to automate curriculum development or assessment methodically by proving it against the hierarchical constructs of Bloom's Taxonomy, education processes can degenerate into mere information retrieval as opposed to real cognitive progression. Therefore, the assessment of how LLMs match with the Bloom's Taxonomy is not only a technical issue, but a pedagogical necessity, which has a direct implication on the quality of instruction and academic honesty. [9].

B. CURRENT CHALLENGES IN AI-DRIVEN PEDAGOGY

Regardless of the huge potential of automated curriculum design and assessment, the application of LLMs to educa-

tional settings is limited to several interconnected technical, pedagogical, and ethical issues. Such restrictions especially surface when the LLMs are compared to other educational systems like Bloom's Taxonomy where precision, consistency and cognitive intent play a very important role. The solution to such challenges is thus a precondition of responsible and systemic implementation of AI-driven pedagogy.

Accuracy and Stochastic Instability: The most significant impediment to the pedagogical use of LLMs is the credibility of the results. LLMs are probabilistic generators of responses, unlike rule-based educational systems, and are trained on distributions and thus generate responses. Although this allows language to be flexible, it also brings with it stochastic instability in the form of the same prompt obtaining uneven or even conflicting responses on different trials. Such variability in the educational setting takes the form of hallucinations, whereby the models are sure to err in classifying the learning goals or creating tests that do not match the desired cognitive level. [10]. Even small errors of classification, such as confusing *Understanding* with *Analyzing* are likely to interfere with the teaching sequence, erode scaffolded learning frameworks, and invalidate the results of assessment.

Semantic and Contextual Ambiguity: Educational learning objectives are context-dependent, and cognitive functions of learning objectives are often rooted in disciplinary norms, curriculum and teaching objectives. Faculty members usually approach these goals in a comprehensive way, in which they incorporate these goals into a greater pedagogical narrative. On the other hand, LLMs are heavily reliant on statistical pattern recognition that is biased towards surface-level linguistic features, which impedes the ability to capture semantic nuances within instructional text meaning they often provide stable and grammatically correct responses but in ways that are pedagogically incorrect.

Empirical studies have found that classifications, generated by LLMs, might look correct on the face of it when viewed in isolation; yet, when factoring in the presence of particular cognitive depth, as outlined by the Bloom's Taxonomy, they tend to have little or nothing to do with the given depth specifically. [11]. This gap between the linguistic and pedagogical soundness significantly reduces the faith of the educators in automated classification and generation systems.

Data Provenance and Algorithmic Bias: LLMs have a tight connection between their pedagogical competence and their training data quality, diversity and representativeness. In the educational sphere, such interdependence creates a strong risk of a garbage-in, garbage-out threat. When training corpora with a disproportionate number of lower-order recall or comprehension items, which is common when using large scale, web-scraped data, models tend to show weaknesses in identifying or generating higher-order question requiring assessment or generation. [8]. The identified imbalance can lead to the emergence of algorithmic bias in LLMs models, whereby the models will selectively assign lower-order cognitive labels not because of some structural limitation, but

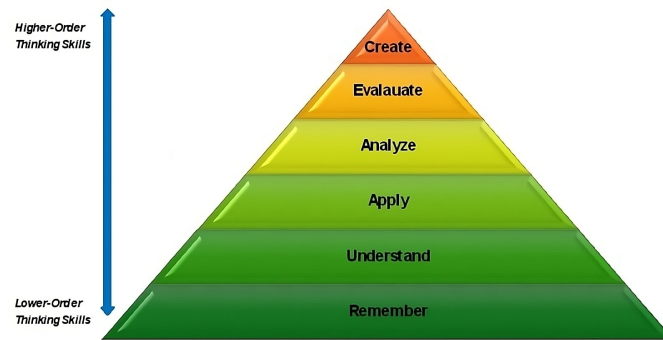


FIGURE 1. Bloom's Taxonomy Levels: The Hierarchy from Lower-Order to Higher-Order Thinking

because the data distributions are skewed. These prejudices are particularly harmful in the matters of curriculum design since they threaten to homogenize the cognitive diversity unintentionally and reduce the higher-order learning goals to the background.

Ethical and Practical Implications: In addition to technical constraints, LLMs have a significant number of ethical and practical implications when used in education. Additional contributing factors to data privacy and academic integrity problems are the effect of cognitive offloading where students and educators are over-reliant on AI systems to perform tasks that should develop critical thinking and meta cognitive processes. This excessive use of automated content generation would run the risk of destroying the ability of learners to analyze, evaluate, and create on their own, which are the key competencies emphasized at the highest levels of Bloom's Taxonomy [4]. Unless artificial pedagogical protections are implemented, the introduction of AI poses a threat to reduce education not into a cognitive process of development but into a form of passive consumerism.

The Empirical Gap: LLMs have often been theorized to have enormous educational potential, yet only a limited amount of empirical work has been conducted, is often fragmented and disproportionate between activities and domains. Much of the previous research focuses on overall ability to generate (as in the case of explanation and feedback generation); however, few studies to date systematically evaluate whether such products correspond with existing pedagogical theories, like the taxonomy proposed by Bloom, in which cognitive intent is explicitly upheld at hierarchical levels. Also, robustness studies on models are usually based on one-model snapshots, most typically ChatGPT, instead of comparing benchmarks across architectures and settings such as discriminative versus generative models, base versus fine-tuned models. Therefore, researchers and educators do not have a converged evidence to define the situations when LLMs can be used in a reliable manner to determine the level of cognitive processes and the situational conditions. This oversight thus provides incentive to a concerted synthesis of empirical results on the applications and constraints of LLMs in the framework of organized pedagogical situations.

[7].

C. FOCUS OF THE STUDY

This SLR explores the level of fidelity of alignment of LLMs to the hierarchical cognitive structure of Bloom's Taxonomy. Although AI applications in education represent a wide range of capabilities, the present review purposefully focuses on two main dimensions directly affecting pedagogical validity, namely: (i) the classification accuracy of the models in recognizing Bloom-oriented cognitive levels, and (ii) the ability of the models to generate pedagogically sound assessment resources that are consistent with the desired cognitive requirements [12]. Focusing on the dimensions mentioned, the research provides an evidence-based framework through which one can evaluate the possibility of having educational tools that are driven by the familiar term of LLMs to facilitate the design of the learning process without jeopardizing the element of cognitive rigor.

1) Evaluating Model Competence

The primary objective of this SLR is to evaluate whether modern LLMs, including generic conversation agents and highly specific, highly refined models, can reliably diagnose the presence of particular cognitive levels (such as between differentiating between the tiers of Understanding and Analyzing), and pedagogizing intent often has manifest implications but is often hidden. This questioning is critical to form a baseline of trust among educators and instructional designers that are interested in introducing these models into automated curriculum auditing, learning-objective tagging, and assessment-alignment processes [12]. In practice, poor classification at this stage can destabilise the sequence of instruction, undermine the effectiveness of scaffolding and shape the desired learning process out of shape.

2) The Data Ecosystem

One of the major secondary concerns of this SLR is the ecosystem of datasets supporting the Bloom-aligned educational modeling. The quality, diversity and representativeness of training and evaluation corpora are the inherent limitations of AI systems regarding their performance and

generalizability to new scenarios and objectives [13]. Current literature suggests that there has always been an academic gap in this area: models are commonly trained and tested on small datasets that have a higher number of lower order recall-style items, which do not reflect higher order tasks that need synthesis, evaluation, and creative constructions to be identified and created [4].

In this regard, this review provides a stringent analysis of the provenance and design characteristics of datasets that are utilized in previous studies, and specifically:

- **Coverage:** How well the datasets cover the full spectrum of Bloom's Taxonomy cognitive levels, and especially the evaluation and create levels which are often poorly represented in datasets, and thus are the least represented levels of cognitive skills [8].
- **Modality:** The heterogeneity in terms of the assessed formats such as multiple-choice items vs. scenario-based and open-response questions, in which the degree to which higher-order reasoning is actually necessitated..
- **Validation:** This validation method may be further defined as an assurance of alignment with ground truth such as expert annotation and inter-rater agreement, a distinction between unverified web-scraped corpora and curated educational datasets can be made at this point in time. [9].

Combining the evidence on the model performance and on the dataset quality, this review aims at defining the best practices, as well as research directions that should be followed to develop next-generation AI instruments which are not only linguistically fluent, but also pedagogically robust and cognitively aligned with the model proposed by Bloom's framework. [4].

D. COMPARISON WITH EXISTING WORKS

The use of Natural Language Processing (NLP) in educational context has been thoroughly studied using questionnaires which concentrate on automated assessment, feedback generation, adaptive learning, and the morality of GenAI. However, even with the increased literature in this area, there is a strong deficiency in systematic reviews that quantitatively estimate the structural alignment between LLMs and Bloom's Taxonomy a model that explicitly provides a cognitive purpose and development in pedagogical design [14]. Therefore, the existing reviews are rather superficial while covering a wide range of applications without focusing on the areas of cognitive validity and instructional fit.

An even more detailed synthesis of the previous literature demonstrates that there are three common methodological limitations that this review directly concerns.

Domain and Model Silos: Current literature typically works under limited experimental boundaries, typically testing one model (most often ChatGPT) in one field of discipline (usually medical or computing education) only once [15]. Current literature often conducts research within limited experimental conditions, typically assessing one model (usually

ChatGPT) in one field of disciplinary application (usually medical or computing education) only once. Although these studies are useful to obtain localized information, they are not suitable in cross-domain generalization and conceal performance variations across model architectures. In turn, the literature does not have a comparative appreciation of the efficacy of different model families, such as fine-tuned discriminative models such as BERT based classifiers versus large generative models such as GPT-4 to different pedagogical tasks and domains.

The Higher-Order Blindspot: Second weakness is with regard to how cognitive complexity is treated. The majority of the previous reviews focus on the basic NLP-assisted activities, including automated marking, short-answer grading, and content synopsis. These activities are often assessed without a clear distinction on the levels of the cognitive hierarchy of Bloom's Taxonomy, thus implicitly assuming learning activities as cognitively homogeneous. Thus, algorithmic issues surrounding HOTS such as analysis, assessment, and composition are either under-researched or confused with superficial understanding [16]. This blind-spot limits the understanding of whether LLMs can be effective in supporting deeper learning and are not just a way of writing plausible text.

Dataset Omission: The most severe weakness of available literature on surveys, perhaps, is the lack of rigorous, systematic analysis of the datasets used to train and benchmark educational LLMs. Without a thorough evaluation of the provenance of datasets, cognitive coverage and validation procedures, it is impossible to determine whether these empirical restrictions within model behaviour are due to the architectural constraints or the existence of biased and incomplete training corpora. This is a critical gap, in particular to Bloom-congruent tasks, where the under-representation of higher-order cognitive goals can mean the development of a systematic bias favoring models to lower-order classification patterns.

Table 1 Combines these constraints and puts the current review into perspective with the best previous studies. Unlike other surveys, this systematic literature review assumes a cross-domain, architecture-conscious and data-driven approach. Providing a clear separation between classification and generation activities, analyzing the properties of an audit dataset, and assessing the performance at all six levels of the blooms taxonomy cognitive domain, this review offers a pedagogically-based synthesis, which is currently lacking in the literature.

E. RESEARCH QUESTIONS

This SLR is guided by a set of focused research questions designed to examine the pedagogical alignment of LLMs with Bloom's Taxonomy from architectural, empirical, and data-centric perspectives.

TABLE 1. Comparison with Existing Studies and Research Gaps Addressed by This SLR

Study	Scope & Focus	Documented Limitations	Verified Contribution of This SLR
Lan et al. (2024) [14]	General survey of NLP applications in education (automated scoring, feedback, content generation)	Lacks alignment with Bloom's cognitive hierarchy; treats question generation and assessment tasks as cognitively uniform	Provides a structured evaluation of LLMs performance across all six Bloom's levels, explicitly identifying systematic underperformance at higher-order levels (<i>Analyze, Evaluate, Create</i>)
Shahzad et al. (2025) [16]	Review of challenges in Generative AI adoption (ethics, privacy, institutional readiness)	Does not include comparative benchmarking of model architectures or task-specific performance evaluation	Introduces an architecture-aware synthesis distinguishing discriminative (BERT) and generative (GPT-family) models, contextualized by task type (classification vs. generation)
Herrmann et al. (2024) [15]	Empirical evaluation of ChatGPT on medical exam questions using Bloom's Taxonomy	Restricted to a single domain (medicine) and a single model; does not generalize findings across disciplines or tasks	Extends Bloom-based evaluation across multiple domains (medicine, engineering, business), enabling cross-domain comparison of higher-order reasoning performance
Baidoo-Anu & Owusu Ansah (2023) [17]	Review of ChatGPT use in higher education (writing support, tutoring, assessment design)	Does not map observed capabilities to cognitive levels or analyze limitations in HOTS	Explicitly targets the documented "Higher-Order Gap" by separating lower-order and higher-order task performance within Bloom's framework
Zhang et al. (2021) [18]	Early BERT-based Bloom's level classification in computing education	Single domain, single task (classification), and pre-LLM architecture scope	Situates early BERT-based results within a longitudinal synthesis (2021–2025), contrasting classification maturity with ongoing generation challenges
Li et al. (2022) [19]	Large-scale CLO classification benchmark using Bloom's Taxonomy	Focuses exclusively on classification; does not address content generation or general-purpose LLMs	Integrates classification benchmarks with generation-focused studies, demonstrating divergence between solved classification tasks and unresolved generation tasks
This SLR	Comprehensive synthesis of Bloom's Taxonomy alignment with LLMs (2018–2025)	—	Distinct Contributions: (1) Separation of classification vs. generation tasks (2) Cross-domain synthesis beyond single-discipline studies (3) Architecture-sensitive comparison (discriminative vs generative models) (4) Identification of persistent higher-order cognitive limitations (5) Evidence-based framing for pedagogically safe LLM integration

TABLE 2. Research Questions (RQs)

RQ	Research Question
RQ1	How effective are LLMs in classifying learning outcomes at each level of Bloom's cognitive framework?
RQ2	Which educational datasets are most effective for training and evaluating LLMs for Bloom's Taxonomy classification?
RQ3	How do different LLMs compare in classifying learning objectives according to Bloom's Taxonomy?
RQ4	What are the limitations and opportunities for improving LLMs-based Bloom's Taxonomy classification?

F. RESEARCH OBJECTIVES

To systematically address the above research questions, this review pursues the following research objectives:

RO1. To synthesize existing empirical evidence on the use of Large Language Models for Bloom's Taxonomy-aligned educational tasks.

RO2. To examine how architectural design influences the capabilities of Large Language Models in Bloom-level classification tasks.

RO3. To analyze how educational datasets and benchmarking practices influence Bloom's Taxonomy classification outcomes in LLM-based systems.

RO4. To identify systemic limitations, risks, and future research directions for improving Bloom-aligned applications

of LLMs in education.

G. SIGNIFICANCE OF THE STUDY

The study goes beyond a technical assessment of the effectiveness of the models, and an evidence-based synthesis will outline the ways AI systems can be properly integrated into educational practice. The review provides systematic insights, which are relevant to the practice of pedagogy, institutional decision-making, and subsequent research on educational AI through a systematic analysis of the concordance between LLMs and the Bloom's Taxonomy.

1) Pedagogical Impact: From Automation to Augmentation

Regarding educators and instructional designers, the current review explains the conditions under which LLMs can be used as a supplement instead of substitutes to pedagogical judgment. By distinguishing between the tasks when automated classification proves to be a reliable tool and the tasks that demand higher-order reasoning, the study helps to implement AI in administrative processes with a more deliberate approach, including the tagging of learning objectives and the audit of curricular content. Such selective application reduces the instructional overhead as well as protecting agency of educators when making cognitively challenging design choices.

In the learner-centered view, the specific correspondence between the AI-generated content and the Bloom's cognitive taxonomy is a precondition of the meaningful personalization. In the instances (such as Understanding versus Analyzing) where LLMs models have sufficient ability to make reliable discrimination between adjacent levels, they can be used to support adaptive scaffolding by changing problem difficulty in proportion to learner performance. The alignment is especially relevant in heterogeneous classrooms, where the differences in the instructional pacing and cognitive readiness make the delivery of the homogenous content rather ineffective..

2) Strategic Guidance for Policymakers

At the institutional scale, this review provides a logical model of reviewing AI tools that goes beyond the surface functionality. The results, through careful recording of randomness in model performance through the changing levels of cognition and the type of task, provide supporting evidence in making decisions grounded in evidence with regard to adopting technology and making purchase decisions. Instead of taking a wholesale position in support or opposed to the use of AI systems, educational leaders can use these insights to outline validation standards, boundaries of application, and quality assurance standards that align with the pedagogical goals and support academic integrity.

3) Foundations for Future Research

Finally, this review determines a structural gap in modern learning AI research connected with dataset design and evaluation. Through the synthesizing empirical evidence that shows presence of a nexus between model behavior and dataset coverage as well as annotation faithfulness, the study shifts the emphasis on model-centric model optimization to data-centric research methodologies. Such a perspective provides a foundation on future research in domain-specific, cognitively balanced and balanced education corpora, as well as, more rigorous benchmarking processes, thus contributing to the development of AI systems informed by pedagogy and not solely linguistically competent.

H. STRUCTURE OF THE PAPER

This paper is organized to ensure a clear progression from conceptual grounding to empirical synthesis and interpretation.

- **Section I (Introduction):** Presents the research context, scope, and guiding research questions.
- **Section II (Methodology):** Describes the systematic literature review protocol following PRISMA guidelines, including search strategy, selection criteria, and quality assessment.
- **Section III (Results):** Reports synthesized findings on Bloom-aligned classification and generation performance, datasets, and model architectures.
- **Section IV (Discussion):** Interprets results in light of pedagogical theory, technical constraints, and implications for educational practice and policy.
- **Section V (Conclusion and Future Directions):** Summarizes key findings and outlines directions for future research.

II. METHODOLOGY

A. SEARCH STRATEGY

In order to ensure that the synthesis of the literature was rigorous, transparent and reproducible, this review followed a systematic search protocol according to the PRISMA guidelines. The search strategy was carefully designed to balance sensitivity in the sense of maximizing the number of relevant studies retrieved and specificity in the sense of eliminating the wider NLP literature that does not have a clear pedagogical basis. Based on this, the protocol combined selective database selection with carefully formulated Boolean queries that were selected to retrieve studies that fell into the intersection of LLMs and Bloom's Taxonomy.

1) Data Sources and Search Venues

Since this review has an interdisciplinary nature (including computational linguistics, educational technology, and learning sciences) the sources were chosen in a stratified way, covering complementary bibliography areas to cover everything.

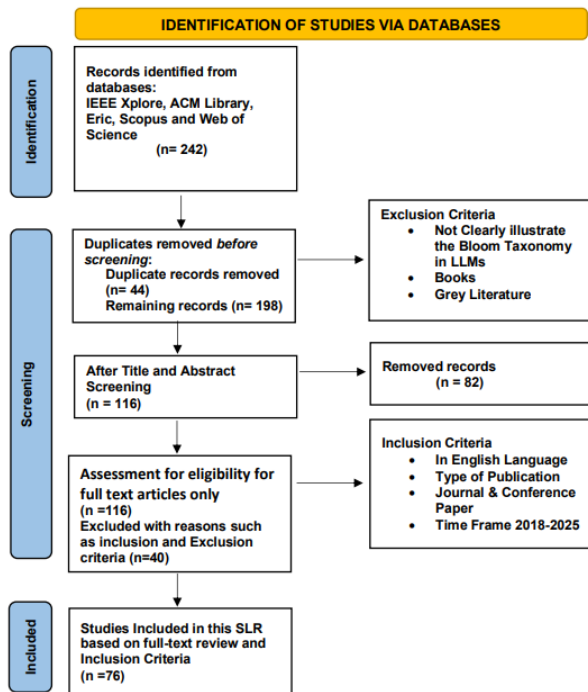


FIGURE 2. PRISMA flow diagram illustrating the study identification, screening, and selection process.

- **Technical Core (IEEE Xplore and ACM Digital Library):** The databases were selected to search systematically the peer-reviewed literature on the algorithm design, model architecture, and the real application of machine learning systems. IEEE Xplore covers the field of engineering-based AI applications in detail, and the ACM Digital Library includes the basis of studies in computational frameworks and human-computer interaction that can be applied to educational technology.
- **Multidisciplinary Indices (Scopus and Web of Science):** We have included Scopus and Web of Science (WoS) to make sure that the disciplines are well represented and that the evidence standards are high. The choice of Scopus was made because it is a comprehensive indexing database for educational technology and cognitive science research
- **Pedagogical Specialization (ERIC):** In order to reduce the technical bias that is usually inherent in engineering based databases, the Education Resources Information Center (ERIC) was consulted. ERIC was necessary in obtaining pedagogy-oriented research, curriculum research, assessment modeling, and applied research on the topic of Bloom-taxonomy across the K-12 and higher-education settings.

2) Search Terms and Query Design

To define the semantic scope of the systematic review, search terms were grouped into two conceptual clusters, one of which was pedagogical context and the other conceptual

cluster was the computational intervention. The last queries were developed based on the Boolean logic (*Cluster A AND Cluster B*) to identify the studies that directly interact with both the theory of education and AI-based modeling.

• Cluster A: Pedagogical Frameworks

- “*Bloom’s Taxonomy*”: Used as the primary anchor word to find the studies that discussed hierarchical cognitive modeling.
- “*Taxonomy of Learning Objectives*”: To include those studies that use more general or different educational terms.

• Cluster B: Computational Architectures

- “*Large Language Models*” (LLMs): The overarching technical category.
- “*Generative AI*” and “*Transformer Models*”: There are also included in order to incorporate both research done before the development of LLMs and foundational work on transformer architectures.
- *Model-specific identifiers* such as ‘ChatGPT’, ‘Gemini’, ‘LLaMA’, ‘BERT’: The explicit search was done to improve the sensitivity of search since published applied studies often mention specific tools but do not index the generic term of ‘LLM’.

The dual-cluster design helped to include both conceptual analyses that refer to the AI systems that align with Bloom-based evaluation and empirical analyses that relate to the model implementations performed.

3) Search String Formulation

As a way of operationalizing the retrieval protocol, a composite Boolean search query was created using logical operators and wildcard truncation. The design was developed to cover the overlap of three semantic dimensions:

- (1) Pedagogical framework,
- (2) Educational context or outcome,
- (3) AI architecture.

The search query is finalized in Table-3. The truncation operator, which is asterisk (*) was used to manage morphological variations as “model” versus “models”).

4) Temporal Scope

The search was limited to studies published between January 2018 and November 2025. This period was selected to capture the full developmental trajectory of transformer-based language models in educational contexts.

- **Start period (January 2018):** Marks the introduction of transformer-based architectures, including BERT and GPT-1, representing a paradigm shift from recurrent neural networks.
- **End period (November 2025):** Ensures inclusion of recent advances in reasoning-oriented models and their educational applications in the post-ChatGPT era.

TABLE 3. Final Boolean Search Query

Concept	Search String
Concept 1: Pedagogy	"Bloom* Taxonomy" OR "Taxonomy of Learning Objective*"
AND	AND
Concept 2: Context	"application*" OR "impact*" OR "effectiveness" OR "cognitive impact*" OR "learning outcome*" OR "integration"
AND	AND
Concept 3: AI Technology	"Large Language Model*" OR "LLM*" OR "GPT*" OR "Bard" OR "Gemini" OR "Transformer Model*" OR "LLAMA" OR "ChatGPT*"

5) Protocol Rationale

The search protocol was elaborated in a manner that highlight the process of recall and not precision. The combination of pedagogical constructs based on the Taxonomies of Bloom with a set of explicit terms related to AI architecture systematically cuts the generic NLP research and retains the studies where empirical appraisals of educational relevance are offered. Consequently, the corpus that has been created is an illustration of a theoretical interaction with cognitive paradigms and empirical data concerning model behavior in the education environment.

B. INCLUSION AND EXCLUSION CRITERIA

In order to make sure that the synthesized evidence was both methodologically rigorous and directly oriented towards the research objectives, a series of eligibility criteria was predetermined. All studies that were retrieved were filtered through specifically defined Inclusion Criteria (IC) and Exclusion Criteria (EC) that are outlined in Table 4. Such organized screening had the benefit of allowing screening to be consistent between stages, and reducing the possibility of subjective selection bias.

The eligibility criteria were formulated to give preference to studies in the intersection of pedagogical theory, that is, Bloom's Taxonomy, and then current LLMs architectures, and to systematically weed out general NLP research with no explicit edification context or taxonomic foundation.

C. STUDY SELECTION PROCESS

Study selection followed the PRISMA guidelines to ensure transparency and reproducibility Figure. 2. The screening process was carried out in three stages: Initial identification and deduplication of records, title and abstract screening based on the inclusion criteria, and full-text assessment for eligibility. Studies lacking empirical evidence or explicit alignment with Bloom's Taxonomy were excluded, resulting in a final set of 76 studies included for data extraction.

D. DATA EXTRACTION PROTOCOL

To ensure uniformity and systematic synthesis of all the studies included, a systematic data extraction tool was created. The variables that were extracted were categorized into three different categories that directly related to the respective research questions.

- **Bibliometric Information:** Title of the publication, authors, year of publication and the venue (journal or conference).
- **Methodological Characteristics:** The LLMs architecture such GPT-4, BERT used, the training or fine-tuning data (i.e., proprietary and open-source), and the sample sizes that are mentioned.
- **Pedagogical Attributes:** Specified Bloom's Taxonomy levels (lower versus higher order), area of application such as STEM education, medical education and output measures such as accuracy, F1-score.

E. QUALITY ASSESSMENT

The quality appraisal was conducted in order to mitigate the risk of bias and assure that the synthesized findings were valid. Every study was rated against five Quality Assessment Criteria (QACs) on a three-point basis:

- 0 = Failed to meet criterion (Unsatisfactory)
- 0.5 = Does not meet criterion (Partial)
- 1 = Completely satisfied criterion (Satisfactory)

A total of 5 points was the maximum possible score. Evidentiary standards were upheld by eliminating studies that had a minimum threshold (Score < 2.5) in the aggregate synthesis as stated in the Table 5.

III. RESULTS AND FINDINGS

This review is based on empirical studies published since 2018 and focuses on the application of the LLMs in the assessment of learning and the compatibility of the learning goals with the Bloom's Taxonomy. The reviewed studies were analyzed using both structured data extraction and narrative synthesis, thus representing the variability in methodological design, detail of reporting, and practice of evaluating the studies in the literature corpus.

The literature clearly shows that academic interest has grown recently, in the aftermath of 2022, when the use of large-scale generative LLMs in education started to proliferate. Historical research has mostly considered exploratory assessments and proof-of-concept implementations, but more recent research is conducting more systematic examinations on task-specific uses of LLMs in classification, generation, and assessment-related tasks.

In the studies that were reviewed, methodological and performance patterns did have a consistent variance across

TABLE 4. Summary of Inclusion and Exclusion Criteria

Inclusion Criteria (IC)		Exclusion Criteria (EC)	
Code	Description	Code	Description
IC1	Pedagogical Alignment: Explicit use of Bloom's Taxonomy (original or revised) as the primary framework for classifying learning objectives or evaluating cognitive levels.	EC1	Thematic Mismatch: General NLP studies such as sentiment analysis, code generation without educational application or taxonomic mapping.
IC2	Technological Scope: Implementation or evaluation of Transformer-based language models such as GPT-4, BERT, LLaMA, Gemini.	EC2	Superficial Reference: Bloom's Taxonomy mentioned only nominally without integration into methodology or evaluation.
IC3	Publication Type: Peer-reviewed journal articles or full archival conference papers.	EC3	Non-Archival Sources: Editorials, posters, keynote abstracts, and non-peer-reviewed reports.
IC4	Temporal Scope (2018–2025): Studies published in the post-Transformer era to ensure relevance to contemporary LLM architectures.	EC4	Architectural Obsolescence: Studies relying exclusively on pre-Transformer models such as, RNNs, LSTMs.
IC5	Language and Accessibility: Full-text availability in English via institutional or publisher repositories.	EC5	Redundancy or Inaccessibility: Duplicate publications or studies without accessible full text.

TABLE 5. Quality Assessment Criteria (QACs)

ID	Assessment Metric	Score = 0 (Unsatisfactory)	Score = 0.5 (Partial)	Score = 1 (Satisfactory)
QAC1	Research Design Clarity: Are the research objectives clearly stated?	No, objectives are not stated	Partially, objectives are stated but not clear	Yes, objectives are clearly stated
QAC2	Methodological Reproducibility: Is the research methodology well defined?	No, the methodology is not well defined	Partially, the methodology is defined but steps are missing	Yes, the methodology is well-defined
QAC3	Taxonomic Validity: Does the study clearly discuss the impact of LLM on Bloom's Taxonomy?	No, the study does not clearly discuss the impact	Partially, the study discusses but unclearly	Yes, the study clearly discusses the impact
QAC4	Strategic Analysis: Does the study properly discuss the strategies used for LLM on Bloom's Taxonomy?	No, strategies are not discussed	Partially, strategies are discussed but unclearly	Yes, strategies are adequately discussed
QAC5	Technical Depth: Does the study properly discuss the techniques used by LLM on Bloom's Taxonomy?	No, techniques are not discussed	Partially, techniques are discussed but unclearly	Yes, techniques are appropriately discussed

Note: Maximum possible score: 5 points. Inclusion threshold: Studies scoring below 2.5 were excluded from the final synthesis.

three main dimensions (type of task i.e. classification verses generation) and cognitive complexity as outlined in the Bloom's Taxonomy (lower-order verses higher-order thinking abilities) and discipline (such as medicine, engineering, business, and general education). The results published in the sections that follow have these dimensions as the organising structure.

A. OVERVIEW OF THE RESEARCH LANDSCAPE

1) Methodological Approaches

The reviewed articles used a wide range of approaches, as to which the methods could be classified into five predominant groups. All these methods demonstrate how the Blooms Taxonomy has been applied to operationalize the performance of the LLM under controlled, comparative, and applied learning environments.

Experimental and Controlled Designs. A number of studies used controlled experimental designs in order to determine the effectiveness of instruction or evaluation using LLMs compared to traditional pedagogical methods. For example, Al Ghazali et al. [20] Relative to the effectiveness of ChatGPT guided versus teacher guided teaching in high school chemistry, measuring the learning outcomes in Bloom compatible cognitive domains. In medical education, Ambadasu et al. [21], Westerlund et al. [22], and Jawanpuria et al. [23] The current paper tested ChatGPT generated replies to strictly tested exam questions, and the results were compared to the performance of students and professional reviewers. Taken together, these studies support the feasibility of the empirical assessment of LLMs using the taxonomic scheme proposed by Bloom in the context of the instruction under bona fide conditions.

Comparative Performance Studies. A major part of the literature has concentrated on comparative assessments of a variety of LLMs. Bielówka et al. [11] analyzed ChatGPT-3.5 on a national medical exam by classifying the questions based on the Bloom's Taxonomy and using non-parametric statistical methods.. Hultberg et al. [7] and Bajcetić et al. [12] The strategy was expanded by a comparative evaluation of various chatbot systems in economics and medical education areas respectively. More comparative studies were done by Shojaae-Mend et al. [8], Joshi et al. [24], and Turunç et al. [25] also evaluated the differences in accuracy, the quality of the reasoning, and adherence to the higher-level cognitive requirements. Taken together, these experiments indicate systematic changes in model designs as opposed to consistent performance.

Framework-Oriented Integration Studies. Such a line of enquiry has proposed pedagogical models of how to incorporate LLMs in the instructional design. Newham et al. [6] introduced a new Constructive Alignment-based system of flipped learning driven by ChatGPT generated materials, while Mandai et al. [4] theoretically mapped LLM capabilities to Bloom's revised taxonomy and reflective learning models. Other implemented frameworks especially the SACITED improvement by Salcedo-Lagos et al. [26], the Higher-Order Prompting (HOP) framework [27], and the FPGA model [28]. The empirical studies focus on the systematic use of prompts and structured processes as tools of maintaining cognitive rigor. These studies demonstrate how there is a growing trend in formalizing AI-assisted pedagogical methods and abandoning ad hoc implementation.

Prompt Engineering and Question Generation Research. An extensive body of evidence has investigated quick engineering practices and question generation automation. Inthrani et al. [29] Suggested Instructions to Improve the Production of Bloom Compatible Questions in Medical Education. while Kwan et al. [30] and Snehal et al. [31] The research compared assessments generated by the LLM in both situations in engineering and the general education setting. Few-shot prompting, retrieval-augmented generation, and hybrid deep learning pipelines were some of the other techniques studied. [13], [32]. These studies evaluated the content that was produced systematically based on Bloom's Taxonomy as the cognitive measure of importance.

Statistical and Quantitative Evaluation Methods. Lastly, several studies have used strict statistical methods to confirm the classification and generation results in line with the Blooms taxonomy. like that of Krippendorff alpha measures were used. [33], Item Response Theory [10], hierarchical linear modeling [34], and comparative difficulty analysis [35] These methodological strategies were used to test reliability, inter-rater agreement and discriminant validity. The results of the analyses provide quantitative data showing that the performance of LLMs has a systematic variation among the hierarchical Bloom's Taxonomy, with strong differences being present between lower-order and higher-order cognitive tasks.

B. RQ1: EFFECTIVENESS OF LLMs IN CLASSIFYING AND GENERATING COURSE LEARNING OBJECTIVES BASED ON BLOOM'S TAXONOMY

Throughout the literature reviewed, there is consistent evidence of two functional capabilities of Bloom-aligned tasks, (i) classifying learning objectives or assessment items based on levels of hierarchy of Bloom's Taxonomy and (ii) generating educational materials that meet the questions, explanation and assessment materials), etc. It has been found that there are inequality in performance patterns between these modalities and these are dependent on type of model (discriminative or generative), cognitive domain (lower or higher) and context of discipline.

1) Classification Effectiveness

: Between Zero-Shot Limitation and Fine-Tuned Mastery. Although the limited available empirical data on zero-shot or slightly adapted generalist LLMs show that classification performance is highly task-dependent across task difficulty levels. Bajcetic et al. [12] The research article indicated that GPT 4, Bard and Claude were able to achieve an approximate of 90% accuracy in the classification of lower-level orders of medical multiple-choice questions, including those that belonged to Remember and Understand levels. Nevertheless, the systems were less consistent in drawing the line between the higher-order categories that were closely related like Analyze versus Evaluate, where inter-annotator agreement dropped to 65-72%. Hultberg et al. [7]. In studies we found similar uncertainties between ChatGPT-4 and LLaMA 2 answering questions in economics, with classification uncertainty increasing among those items that require multi-step thinking. In neurophysiology tasks, Shojaae-Mend et al. [8] The findings prove a relatively higher performance in factual recall (mean 3.87/5) with respect to integrative tasks in analysis (mean 2.43/5). In a short-answer medical test, Morjaria and Burns [36] assessed ChatGPT -3.5 and did not find statistically significant differences between the domains of Bloom; the model scored an average of 3.29/5, which is higher than the minimum passing mark of 2.38/5, but lower than the historical average of the class (3.67/5).

Multiple studies attribute these early inconsistencies to the difficulty of disambiguating cognitive intent from short prompts and limited contextual signals. Herrmann et al. [15] highlighted challenges in reliably differentiating higher-order levels when Bloom labels are used as evaluation categories. Sein Hui [2] documented instances where analysis-level questions were classified as comprehension tasks, and Moore et al. [37] reported persistent difficulty distinguishing *Apply* from *Analyze* in the LearningQ dataset when using generalist approaches.

In contrast, studies using fine-tuned encoder-based architectures report substantially higher and more stable classification performance. Li et al. [19] constructed a large-scale benchmark of **21,380 Course Learning Objectives (CLOs) from 5,558 courses** at Monash University and demonstrated that fine-tuned BERT models achieved Cohen's κ of 0.87–

0.93 and F1 scores of 0.88–0.95 across all six Bloom levels, outperforming conventional classifiers (Naive Bayes, Logistic Regression, SVM, Random Forest, XGBoost). Li et al. [19] further reported that binary classifiers (one per cognitive level) slightly outperformed multi-class multi-label setups, suggesting benefits to decomposing Bloom's levels into separable decision boundaries.

Subsequent studies extend these results across domains and model variants. Punyatoya et al. [38] reported that domain-specific fine-tuning on VTU university exam questions increased BERT-based classification accuracy to **98%**. Mazza et al. [39] found fine-tuned RoBERTa achieved 91.66% accuracy, compared with approximately 72% for zero-shot LLM baselines. Earlier work by Zhang et al. [18] reported 82.61% accuracy for frequent levels (*Remember*, *Understand*, *Analyze*) but only 59.2% for infrequent levels (*Apply*, *Evaluate*, *Create*), emphasizing the influence of label distribution and dataset balance. Deroy et al. [40] introduced LegalQ (2,023 multilingual context–question pairs) and showed that domain-specific fine-tuning improved classification reliability in legal education.

Overall, the classification literature indicates that Bloom-aligned labeling for curriculum auditing and CLO classification can reach high reliability when implemented through fine-tuned discriminative models trained on sufficiently large and cognitively balanced datasets.

2) Generation Effectiveness: Higher-Order Limitations and Domain-Dependent Advances

Bloom-aligned content generation imposes additional constraints beyond label assignment. Whereas classification is evaluated against a fixed set of categories, generation requires producing pedagogically appropriate materials that reflect cognitive intent and progression. Wang and Fan [41] synthesized evidence from 51 experiments and reported a large pooled effect size for general learning performance ($g = 0.867$) but a smaller effect for HOTS ($g = 0.457$), indicating that improvements in general outcomes do not translate uniformly to higher-order cognitive objectives.

Lower-order generation performance. Several studies report comparatively stronger performance for generation tasks targeting foundational levels. Linnebur et al. [42] observed that GPT-4 maintained above 80% accuracy for biology recall-oriented content but degraded for analysis-level item generation. Su et al. [43] reported ChatGPT-4 achieved 80.75% overall accuracy on Taiwan's Nursing Licensing Exam, with variation by subject such as General Medicine 88.75%, Basic Nursing 63%. They further found that 14.25% of responses contained inconsistencies between answers and explanations, reducing effective accuracy to 74%. In engineering statistics, Kwan et al. [30] found that generated assessments concentrated on application-oriented questions and did not produce *Evaluate* or *Create*-level items. Hwang et al. [13] reported that GPT-4 generated usable MCQs (82.67%), but alignment accuracy declined for *Analyze* and *Evaluate* tasks (below 60%).

Reasoning-oriented models and structured domains.

Recent studies evaluate newer reasoning-oriented systems on cognitively demanding tasks, with performance gains reported in some structured settings. Altermatt et al. [44] benchmarked GPT-o1 on the Chilean Anesthesiology Certification Exam (CONACEM) and reported 88.7% accuracy, with Deepseek-R1 at 86.3%, exceeding GPT-4o (83.9%) and LLaMA 3 70B (60.5%). In software engineering, Guerra and Ernst et al. [45] reported GPT-4 Turbo achieved 100% accuracy in generating architectural design structures, corresponding to Bloom's *Create* level in a rule-governed domain. At the same time, evidence remains mixed: Habiballa et al. [46] reported that in theoretical computer science tasks such as finite automata design, human experts outperformed AI systems, and ChatGPT failed to detect incorrect higher-order solutions in 40% of cases. Nikolovski et al. [47] reported that fine-tuned VectorRAG agents achieved alignment scores not statistically distinguishable from experts ($p \approx 0.12$) in data science contexts, but required substantial domain customization.

Domain-specific performance patterns. Evidence indicates that generation outcomes vary substantially by discipline and task structure.

Medical education: Ambadasu et al. [21] found ChatGPT-4 exceeded student performance in preclinical courses (Cell Biology mean 4.1/5) but showed lower performance in clinical reasoning tasks (mean 3.2/5). Bielówka et al. [11] reported ChatGPT-3.5 achieved 62.7% accuracy on Polish allergology specialist exams, with negative correlation between performance and difficulty (Spearman $r = -0.42$, $p < 0.001$). Jawanpuria et al. [23] reported strong performance in infection prevention and control (4.33/5), while Turunç et al. [25] observed high accuracy in surgical recall but limited capability in complex case analysis requiring synthesis of multiple factors.

Engineering and computer science: Li et al. [48] reported AI-assisted learning improved programming performance at *Apply* and *Create* levels, with smaller impact on *Analyze* and *Evaluate*. Westerlund and Shcherbakov [49] explored LLM integration in coding workbooks aligned with Bloom's Taxonomy, demonstrating how prompt engineering supports various learning levels in programming education. Chenna et al. [50] reported LLM-generated code explanations increased comprehension scores by 23% for *Understand* tasks but only 8% for *Evaluate*-level tasks.

Business and economics: Sorin et al. [51] described an Expertise Paradox in which strength on routine tasks may reduce engagement in higher-order decision-making. Essien et al. [52] reported postgraduate business students improved on lower-order tasks but not on synthesis and evaluation when using ChatGPT. Valcea et al. [53] found ChatGPT generated case study scenarios but struggled with strategic analysis requiring industry expertise. Hu et al. [54] reported Chain-of-Thought prompting improved accounting case analysis but remained below expert judgments in professional skepticism. Liu et al. [55] found ChatGPT-

generated spreadsheet formulas were syntactically correct yet logically flawed for constrained optimization tasks.

STEM disciplines: Al Ghazali et al. [20] reported that ChatGPT-guided chemistry students performed comparably on knowledge-based assessments, whereas teacher-led instruction outperformed on reasoning tasks. Sánchez-Rodríguez et al. [56] reported AI-supported sustainability analysis yielded measurable higher-order gains in 19% of the cohort. Sun et al. [57] found LLMs struggled with scientific reasoning and experimental evaluation except that GPT-4 and Claude showed moderate competence in hypothesis generation but weak experimental design performance.

Language and literacy: Zhu et al. [58] found LLM-assisted writing improved fluency (*Apply*) but did not significantly improve critical analysis (*Analyze*, *Evaluate*). Quille et al. [59] reported GPT-4 achieved the highest scores on machine learning assessments, with stronger performance on lower- and mid-level questions than on open-ended creative responses. Zhang et al. [60] found LLMs were effective for grammar exercises but struggled to generate literary analysis prompts requiring interpretation and evaluation. Rodrigues et al. [61] reported GPT-4 responses to open-ended comprehension questions varied across Bloom levels, with coherent *Understand*-level responses but comparatively superficial *Evaluate*-level analysis. Başaran et al. [62] reported instability in PISA reading tasks, where GPT-4 accuracy varied from 100% to 0% across repeated trials on identical level-6 difficulty items.

Finally, Gomez-Jaramillo et al. [63] reported that AI-mediated programming tutoring improved practical coding (*Apply/Create*) with effect size $d = 0.73$, while impact on abstract algorithm design (*Evaluate*) was smaller ($d = 0.21$), reflecting differential outcomes across task structure and cognitive level.

C. RQ2: WHICH EDUCATIONAL DATASETS ARE MOST EFFECTIVE FOR TRAINING AND EVALUATING LLMS FOR BLOOM'S TAXONOMY CLASSIFICATION?

Across the reviewed studies, dataset design and scale vary substantially, ranging from small pilot collections (often $n < 200$) to large institutional corpora such as $n > 20,000$) and high-stakes standardized benchmarks. To synthesize how evidence is generated and evaluated, the datasets are grouped into five recurring types. These categories differ in purpose (training vs. evaluation), ground-truth strength (expert-labeled vs. implicit), and their ability to represent higher-order cognitive levels.

1) Standardized and High-Stakes Benchmarks

High-stakes professional and standardized examinations provide externally validated content with strong ecological validity and clear performance implications.

Medical licensing and professional exams: Bielówka et al. [11] evaluated ChatGPT-3.5 on 118 MCQs from the Polish National Specialist Examination (PES) in allergology, with manual Bloom-level categorization. Kufel et al. [10] used 117

	Typical Scale	Bloom Coverage	Validation Strength	Primary Use	Frequency in Literature (n=76)
Standardized/High-Stakes Benchmarks	20-1,000+ items	Constrained by exam design	High (external)	Evaluation/Benchmarking	23 studies (30%)
Curated Educational Corpora	100-21,000+ items	Variable; often imbalanced	High (expert annotation)	Training & Evaluation	19 studies (25%)
Student-Generated/Survey-Based	25-200 participants	Limited; task-dependent	Moderate (subjective)	Ecological Evaluation	15 studies (20%)
AI-Generated Datasets	Variable	Intentional but unvalidated	Low (circular risk)	Prototyping/Augmentation	11 studies (14%)
Conceptual/Theoretical Frameworks	N/A	Comprehensive (theoretical)	N/A	Operational Framing	8 studies (11%)

LEGEND: ■ Strong/High ■ Moderate ■ Limited ■ Weak/Low ■ Not Applicable

FIGURE 3. Datasets Characteristic Matrix.

PES questions in nuclear medicine after excluding outdated or incompatible items. Su et al. [43] used 400 questions from Taiwan's Nursing Licensing Exam across five subjects. Altermatt et al. [44] benchmarked reasoning-oriented models on CONACEM (Chilean Anesthesiology Certification Exam), representing a higher-complexity professional reasoning benchmark beyond standard MCQs. Ambadasu et al. [21] and Westerlund et al. [22] evaluated ChatGPT-4 using 304 MCQs across six medical courses, enabling comparison against first-year and third-year medical students. Turun et al. [25] incorporated feedback from 50 medical students and surgical trainees in surgical case discussion settings. Morjaria and Burns [36] collected 40 medical assessment problems (30 submitted to ChatGPT-3.5 and 10 comprising prior student responses), stratified by Bloom levels.

International standardized assessments: Basaran et al. [62] used PISA reading items (2000 and 2009 waves) spanning difficulty Levels 1–6 and evaluated repeated-trial stability on higher-difficulty items.

Engineering and computer science exams: Punyatoya et al. [38] validated classification using VTU exam questions. Kwan et al. [30] generated and evaluated assessment scripts for Probability and Engineering Statistics with Bloom-level analysis, while Zhang et al. [18] used computing education questions for early BERT-based Bloom classification.

Domain-specific question banks: Shojae-Mend et al. [8] designed a 20-item neurophysiology question set spanning Bloom levels, illustrating the use of small but cognitively targeted benchmarks.

2) Curated Educational Datasets

Curated datasets are commonly constructed to support controlled benchmarking, fine-grained Bloom-level coverage analysis, and supervised training/fine-tuning.

The Monash benchmark: Li et al. [19] compiled a large-scale dataset of **21,380 Course Learning Objectives (CLOs) from 5,558 courses** across Monash University. CLOs were manually labeled across the six Bloom's levels (Cohen's $\kappa = 0.80$) and included multi-label instances such as 2,325 with two levels. The dataset provides detailed level distribution: *Remember* (886), *Understand* (5,079), *Apply* (5,074), *Analyze* (2,311), *Evaluate* (2,468), and *Create* (2,955), with 2,607 multi-label instances.

Specialized domain corpora: Deroy et al. [40] introduced LegalQ (2,023 multilingual context–question pairs). Muhamad et al. [64] compiled 146 computer science CLOs to benchmark GPT-4o classification efficiency. Linnebur et al. [42] curated 300 introductory biology MCQs to examine model behaviors including self-correction. Moore et al. [?] incorporated 143 student-generated short-answer questions alongside LearningQ and a Bloom reference set of 600 questions. ZhenTing et al. [33] used TREC collections (QA Track 2002, Robust Track 2003, Terabyte Track 2004), categorizing topics into cognitive complexity levels.

Subject-specific collections: Li et al. [48] used standardized educational questions from K–12 curricula and online platforms across chemistry, mathematics, and coding. Hwang et al. [13] used EduQG supplemented with Bloom-classified examples from OpenStax Biology and Chemistry. Kwan et al. [65] generated ChatGPT-based assessment scripts for engineering statistics categorized by Bloom’s cognitive levels, though the generated content showed bias toward application-level questions with limited higher-order cognitive tasks.

3) Student-Generated and Survey-Based Datasets

Student-generated and survey-based datasets capture authentic classroom interactions and stakeholder perceptions. They are frequently used to study ecological validity and the practical dynamics of Bloom-aligned tasks but may require careful handling due to response variability and subjective labeling.

Student interaction data: Sein Hui et al. [2] analyzed chatbot-mediated learning interactions involving 25 students in an IB Primary Years Programme (PYP) context, interpreting exchanges using Bloom’s Taxonomy. Al Ghazali et al. [20] analyzed outcomes from 39 eleventh-grade chemistry students in ChatGPT-guided versus teacher-led conditions. Weir et al. [66] surveyed undergraduate perceptions of ChatGPT-supported learning activities, reporting higher perceived usefulness for lower-order tasks and concerns about depth at higher-order levels.

Large-scale perception and feedback corpora: Stoyanova et al. [67] analyzed surveys from 87 engineering students and 8 educators to map perceived utility across Bloom’s levels. Zhao et al. [68] constructed a 160,000-word corpus of lecturer feedback on 230 postgraduate assignments to operationalize critical thinking constructs. Essien et al. [52] collected data from 107 postgraduate business students using Bloom-based pre/post testing. Yusof et al. [69] examined ChatGPT-assisted retrieval practice and reported improved exam performance while emphasizing the role of scaffolding. Barcaui [70] conducted a randomized controlled trial and reported reduced long-term retention despite short-term gains. Freire et al. [71] used thematic coding to analyze Bloom-level characteristics in ChatGPT responses to exam questions.

4) AI-Generated Content Datasets

There are several examples of studies that use LLMs to generate datasets, including question banks, thus making it

easier to prototyping faster and achieve predefined coverage goals. Nevertheless, such methods also bring in the aspect of validity, especially with regard to the patterns of artifacts and the uniformity of the generated information.

Synthetic generation frameworks: Shalini et al. [72] proposed prompt-based synthesis explicitly grounded in Bloom definitions to generate assessment items across levels. Snehal et al. [31] fine-tuned a T5-based system using SQuAD1.1, RACE, and DuoRC and produced Bloom-aligned questions. Kwan et al. [30] generated assessment scripts and marking schemes using ChatGPT. Newham et al. [6] used ChatGPT-generated materials across disciplines. Toba et al. [73] developed an LLM question generator combining retrieval-augmented generation and fine-tuning, and reported difficulty in independently producing distinct *Create*-level items (5.7% success rate).

Methodological cautions: Garcia-Ramirez et al. [74] reported ‘easiness bias’ in AI-generated exams, with large differences between lower- and higher-order performance as 98% on *Remember* versus 55% on *Create*, indicating that generated items may lack discriminatory difficulty unless validated.

5) Conceptual and Theoretical Frameworks

Finally, several works contributed conceptual frameworks and operational guidance rather than empirical datasets, especially where higher-order ground truth is difficult to establish using automated metrics.

Pedagogical frameworks: West et al. [27] proposed the HOP framework to structure prompts toward critical engagement. Kwan et al. [75] introduced the FPGA (Flipped Pedagogy and GenAI) model for GenAI-supported flipped classrooms. Mandai et al. [4] provided a theory-based synthesis mapping LLMs capabilities to Bloom’s taxonomy and related educational models. Sorin et al. [51] reported qualitative observations in business education, and Salcedo-Lagos et al. [76] analyzed mathematics teachers’ qualitative feedback on AI-assisted didactic sequence generation.

D. RQ3: HOW DO DIFFERENT LLMS CAN COMPARE IN CLASSIFYING LEARNING OBJECTIVES ACCORDING TO BLOOM’S TAXONOMY?

Synthesis of the reviewed studies indicates that comparative LLMs performance cannot be explained through a simple linear progression of model versions. Instead, performance differences are best characterized by *architectural specialization*, *task type* (classification vs. generation), and *cognitive demand* (lower-order vs. higher-order Bloom levels). Across studies, models cluster into distinct functional tiers rather than forming a single hierarchy.

1) Generalist Generative Models as Performance Baselines
Early comparative studies established baseline performance among general-purpose LLMs. Bajcetić et al. [12] and Hultberg et al. [7] consistently reported that GPT-4 outperformed GPT-3.5 by approximately 10–15% in classification

TABLE 6. Dataset Types Used for Training and/or Evaluating Bloom-Aligned LLM Applications

Dataset Type	Representative Examples	Typical Scale	Primary Use	Bloom Coverage and Common Limitations
<i>Standardized / High-stakes</i>	PISA reading items; CONACEM; PES (Poland); Taiwan Nursing Exam; medical course MCQs; VTU exams	~20–1,000+ items	Evaluation / benchmarking	High ecological validity; cognitive coverage constrained by exam design; higher-order evaluation often exhibits instability under repeated trials [10], [11], [21], [22], [43], [44], [62]
<i>Curated educational corpora</i>	Monash CLOs (21,380) ; LegalQ (2,023); Biology MCQs (300); LearningQ; TREC cognitive tracks	100–20,000+	Training and evaluation	Enables supervised fine-tuning; performance depends on annotation quality and label balance; <i>Evaluate/Create</i> often underrepresented outside large corpora [19], [33], [37], [42], [77]
^r <i>Student-generated / survey-based</i>	Classroom interactions; RCT cohorts; student/educator surveys; lecturer feedback corpora (160k words)	Cohorts: 25–107; corpora up to 160k words	Ecological evaluation	Captures authentic learning behavior; annotations may be subjective; task difficulty uneven; higher-order constructs require expert interpretation [2], [20], [52], [67], [68]
<i>AI-generated datasets</i>	Prompt-synthesized item banks; T5-based question generation; RAG-based synthetic exams	Variable	Training augmentation / prototyping	Allows intentional Bloom-level coverage; risks include style artifacts, inflated performance, and reduced discriminatory difficulty without expert validation [31], [72], [74], [78]
<i>Conceptual / theoretical (no dataset)</i>	HOP; FPGA; theory mappings; qualitative teaching trials	N/A	Operational framing	Provides decision rules for higher-order tasks where automated ground truth is ambiguous; does not yield benchmarkable performance metrics [4], [27], [51], [75], [76]

accuracy, with the strongest gains observed for lower-order tasks (*Remember, Understand*). Hultberg [7] compared GPT-3.5, GPT-4, Bard, and LLaMA 2 on economics questions and found GPT-4 demonstrated stronger comprehension of domain concepts and contextual application. Shojaee-Mend et al. [8] reported that Claude and Bard (PaLM 2) performed competitively on factual recall (mean $\approx 3.5/5$), but GPT-4 achieved higher scores on neurophysiological reasoning tasks (GPT-4: 4.1/5; Claude: 3.7/5; Bard: 3.6/5). Bajcetić et al. [12] further showed that while multiple AI models outperformed first-year medical students in raw accuracy, they exhibited inconsistent classification of higher-order Bloom categories.

Morjaria and Burns [36] reported that ChatGPT-3.5 achieved a mean score of 3.29/5 on medical short-answer assessments, outperforming minimally passing students (2.38/5, $p = 0.008$) but underperforming the historical class average (3.67/5, $p = 0.018$), with no statistically significant differences across Bloom domains. Collectively, these findings position GPT-4 as a strong general-purpose baseline for educational tasks, while also highlighting limitations in higher-order cognitive discrimination.

2) Reasoning-Oriented Models for Higher-Order Cognitive Tasks

More recent studies evaluate models explicitly designed to support multi-step reasoning and structured problem solving. Altermatt et al. [44] reported that GPT-o1 achieved 88.7%

accuracy on the CONACEM anesthesiology certification exam, followed closely by Deepseek-R1 at 86.3%. Both models outperformed general-purpose systems such as GPT-4o (83.9%), GPT-4 (81.2%), Claude 3.5 Sonnet (79.8%), Gemini 1.5 Pro (78.4%), and LLaMA 3 70B (60.5%). These results suggest that reasoning-oriented architectures offer advantages for tasks requiring diagnostic inference and multi-step evaluation.

Evidence from other domains indicates that these gains are **context-dependent**. Guerra and Ernst et al. [45] reported GPT-4 Turbo achieved 100% accuracy in generating software architectural designs, corresponding to Bloom's *Create* level in a structured technical setting. In contrast, Habiballa et al. [46] found that **on the questions that dealt with theoretical computer science and design of finite automata, ChatGPT had a 40% failure rate in spotting incorrect solutions**. On the contrary, human professionals performed better in abstract reasoning tasks. These empirical results indicate that reasoning-based models will have an optimal level of efficacy in areas where explicit logic restrictions are in place and the space of solutions is well defined.

3) Efficiency-Oriented Models for Scalable Educational Workflows

There is also a separate body of research looking at models that have been optimized based on throughput, cost, and usability as opposed to maximising the depth of reasoning.

Muhamad et al. [79] reported, GPT-4o has achieved a classification accuracy of 67.4% on curriculum goals and 75% lower processing costs compared to GPT-4. Extending this work, Muhamad et al. [80] applied GPT-based systems to NCAAA accreditation workflows, reducing report generation time by 87% while maintaining compliance alignment.

However, studies consistently report that efficiency-oriented models require human oversight. Crowther et al. [81] benchmarked GPT-4o, Claude 3.5, and Gemini 1.5 Pro on curriculum design tasks and found that only 38.3% of outputs were usable without editing, with educators spending an average of 12 minutes per learning objective for refinement. Toba et al. [78] reported that AI-generated questions improved student pre-test scores by 15%, but the system independently generated distinct *Create*-level items in only 5.7% of cases. García-Ramírez et al. [74] documented systematic easiness bias in AI-generated exams, with large performance discrepancies between lower-order and higher-order items.

Nikolovski et al. [82] demonstrated that agentic workflows combining VectorRAG with domain-specific prompting achieved alignment scores not statistically distinguishable from human experts ($p \approx 0.12$) in data science curriculum contexts, indicating that architectural orchestration can partially mitigate limitations observed in standalone models.

4) Discriminative Models for Bloom-Level Classification

In contrast to generative systems, discriminative architectures consistently outperform LLMs on Bloom-level classification tasks. Li et al. [19] experimental findings showed that fine-tuned BERT models achieved the Cohen 0.87 to 0.93 and F1 of 0.88 to 0.95 at all 6 levels in the Bloom dataset of Monash CLO. Relative analysis indicated that BERT was significantly better in comparison with Naïve Bayes and Logistic Regression and other more classic classifiers, and higher-order labels were considerably more favored.

These results were supported by later by. Punyatoya et al. [38] experiment claimed an accuracy of 98% on the classification task by fine-tuning BERT on the VTU engineering exam dataset, which is significantly higher than the results of zero-shot LLMs. Mazza et al. [39] found RoBERTa achieved 91.66% accuracy compared to approximately 72% for generalist LLMs. Zhang et al. [18] and Deroy et al. [40] It has been demonstrated that the effectiveness of classification models depends upon the balance of labels as well as fine-tuning studied to a particular domain.

Therefore, the results all demonstrate a functional segregation: discriminative architectures can be effectively used in the task of dependable Bloom-level categorization useful in audit and tagging tasks, whereas generative architectures are used to create new content and construct teaching sequences.

E. RQ4: WHAT ARE THE LIMITATIONS AND OPPORTUNITIES FOR IMPROVING LLM-BASED BLOOM'S TAXONOMY CLASSIFICATION?

Although LLMs can be considered to possess measurable potential in the tasks compatible with the taxonomy proposed by Bloom, the literature reviewed is unanimous about the existence of the continuum of technical, pedagogical, and methodological limitations. These limitations in the analyses are not just limited to early fears of factual hallucinations but to a wider set of systemic problems of output stability, professional judgment, learner agency and the authenticity of the evaluations.

1) Technical Limitations

Probabilistic Volatility and Output Instability. One of the critical technical limitations is the stochastic character of the outcomes of the LLM.. Shojaee-Mend et al. [8] and Bajcetić et al. [12] reported variability in Bloom-level classification across repeated evaluations of identical prompts. Başaran et al. [62] It has provided a very timely example in the context of high-stakes PISA reading tests, where the accuracy of GPT-4 on the same Level 6 items dropped down to 0 percent on a trial-to-trial basis. Su et al. [43] Our study also revealed internal discrepancies between the answers and the answers to be explained in 14.25% of the nursing test questions, and this led to the decrease of the effective accuracy of 80.75% to 74%. Overall, these findings remind of the fact that modern general-purpose LLMs lack the deterministic reliability required when it comes to autonomous summative assessment.

Plausible Unreliability and Professional Judgment Gaps. In addition to explicit factual errors, empirical studies have found a less apparent type of failure mode due to LLMs, in which the linguistic text produced by the model seems fluent and professionally written, but harbors logic or process errors. Ahmed et al. [83] This has been observed in the case of accounting and auditing where ChatGPT gave audit judgments using proper professional language but with no substantive professional skepticism. Syntactically correct and logically incorrect spreadsheet optimization formulas were also reported by Liu et al [84] as those that novice learners often accepted as correct in the engineering context. In medical education, Inthrani et al. [29] and Bielówka et al. [11] found that model performance declined with increasing question difficulty, with plausible but incorrect reasoning emerging in complex diagnostic scenarios. These findings suggest that LLMs limitations increasingly manifest as failures of expert judgment rather than simple misinformation.

Persistent Challenges with Higher-Order Cognitive Tasks. Despite architectural advances, multiple studies report systematic underperformance on Bloom's higher-order levels. Kwan et al. [30] shows that ChatGPT was lacking in a specific inability to create statistics questions of the level of Evaluate or Create, rather, it created tasks that are application-level in nature. Toba et al. [73] recorded an independent

success rate of 5.7% on generating separate items of creative level. García-Ramírez et al. [74] The presence of an easiness bias has been recorded, in which AI generated examinations generated significantly higher performance on recall tasks 98% and significantly lower performance on creative tasks 55%. This dissimilar pattern can most likely be construed as a result of question formulation, and not a manifestation of unequal learning results. Wang and Fan et al. [41] This difference was statistically measured and found that general learning had a large effect size ($g = 0.867$) whereas higher-order thinking had a moderate effect size ($g = 0.457$). Habiballa et al. [46] Besides, it was stated that ChatGPT was not able to recognize incorrect solution in 40% of abstract theoretical computer science problems.

2) Pedagogical Risks

Cognitive Offloading and Reduced Learner Agency. Several studies raise concerns about the pedagogical consequences of LLM-assisted learning. West et al. [27] Defines cognitive offloading, the process of having learners place in AI systems substantive reasoning instead of thinking actively. Stoyanova et al. [67] empirical studies have found that there is an integrity gap between the perceptions of students and educators: students mainly feel that AI can help to streamline the lower-order tasks, whereas educators fear that critical thinking and academic integrity are going to be eroded. Sorin et al. [51] presented this conflict as an expert knowledge paradox, according to which AI knowledge in routine jobs can suppress the acquisition of more advanced professional reasoning. Al Ghazali et al. [20] provided experimental evidence supporting this concern, reporting that teacher-led instruction outperformed ChatGPT-guided learning on reasoning-intensive chemistry tasks.

Academic Integrity and Assessment Validity. Assessment integrity remains a persistent challenge. Quille et al. [59] according to the study, AI-generated responses are hard to identify with the help of traditional plagiarism detection tools. and Morjaria and Burns et al. [36] found that sometimes AI-generated responses were described as student-generated responses. These results highlight the necessity of redesigning the assessment structures and not just using the detection mechanisms.

3) Methodological Limitations

Circular Validation and Metric Insufficiency. There is a consistent methodological issue of using LLLMs to evaluate the out generated by other LLMs. Bajcetić et al. [12] and Sein Hui et al. [2] It has been noted that the traditional measures of performance accuracy, precision and recall do not effectively capture the semantic and cognitive aspects of higher levels of Bloom's Taxonomy. García-Ramírez et al. [74] It was also seen that the high accuracy scores are possibly an indication of the simplicity of the questions and not true cognitive congruence. Zhao et al. [68] and Saber et al. [85] argue for multi-dimensional evaluation frameworks

incorporating reasoning depth, contextual appropriateness, and coherence.

Dataset Bias and Limited Generalizability. Most of studies use institutional or disciplinary settings that are too narrow to be validated externally. Mandai et al. [4] and Quille et al. [86] also observed to help create a systematic underperformance in specialized or underrepresented settings, such as non-English curricula and professional worlds, training data bias and uneven domain representation.

4) Opportunities for Improvement

In spite of these shortcomings, literature provides multiple action pathways that could be implemented to enhance Bloom-based performance of LLMs.

Domain-Specific Fine-Tuning. Fine-tuning on domain-specific datasets consistently improves classification and alignment accuracy. Punyatoya et al. [38] reported 98% classification accuracy using fine-tuned BERT models, while Li et al. [19] demonstrated near-human reliability through large-scale CLO datasets. Deroy et al. [40] similarly validated domain-specific fine-tuning in legal education.

Hybrid Human-AI Workflows. Human-in-the-loop approaches mitigate plausible unreliability. Shojae-Mend et al. [8] and Freire et al. [71] showed that expert oversight improves alignment and interpretability. Crowther et al. [81] and Muhamad et al. [80] demonstrated that human review substantially increases usability while preserving efficiency gains.

Advanced Prompting and System Architectures. Hu et al. [54] reported that Chain-of-Thought prompting improves reasoning performance, while West [27] and Kwan et al. [75] propose structured prompting and flipped pedagogical models to preserve learner agency. Nikolovski et al. [82] and Toba et al. [73] further show that agentic workflows and retrieval-augmented generation outperform standalone prompting in Bloom-aligned tasks.

Assessment Redesign. Rather than focusing on AI detection, several authors advocate for assessment designs emphasizing process, contextual reasoning, and metacognitive reflection. Quille et al. [86] argue that these designs are more reflective of HOTS in human minds and less prone to AI abuse.

IV. DISCUSSION

In this section, the synthesized findings are interpreted in the context of the research questions and placed in the context of the larger literature on AI-supported education and critically analyzed regarding the pedagogical implications of AI-supported education, the methodological implications of AI-supported education, and the practical implications of AI-supported education. Instead of restating the results of empirical studies, the discourse is about how to explain the observed patterns of performance, how to solve the observed contradictions on the use of LLMs to solve educational tasks based on the Bloom's Taxonomy, and what is truly solved, and what is structurally constrained.



FIGURE 4. Conceptual synthesis of technical, pedagogical, and methodological factors shaping the effectiveness and limitations of LLMs in Bloom's Taxonomy-aligned educational tasks.

Importantly, the reviewed evidence indicates that these outcomes do not arise from single factors in isolation, but from the interaction between **model architecture**, **dataset characteristics**, **task formulation**, and **pedagogical context**. To make these interdependencies explicit, Figure 4 presents a conceptual synthesis that situates technical capabilities and limitations within their pedagogical and methodological constraints.

A. REVISITING THE COGNITIVE ALIGNMENT PROBLEM

Throughout the literature reviewed, one can consistently find that the challenge which LLMs face on Bloom-aligned tasks is not simply consistently difficult, but structurally related to the nature of cognitive operation itself. The lower-order tasks (Remember, Understand, Apply) fit into the statistical learning paradigm underlying the vast majority of LLMs since the former are mostly about pattern recognition, surface-level semantic mapping, and recall. [7], [8].

On the contrary, rigorous critical thinking including analysis, evaluation and synthesis requires a thorough understanding of the context, the ability to make judgments and the ability to provide justification of perceived truths. These abilities go far beyond what is known as being able to predict the next most probable word in a sentence. The current performance gap in AI is not only due to a lack of training data but it is also a design flaw. As a result, a structural incongruity between the functioning paradigms of modern AI systems and the complex-thinking abilities of humans is present [41], [46].

This difference can be used to balance up why simply scaling up the model size such as GPT-3.5 to GPT-4 can only produce incremental improvements in higher-order performance, whereas the improvement in architecture such reasoning models and agentic workflows can be much greater, but the improvement again depends on the domain.

B. WHY CLASSIFICATION IS SOLVED BUT GENERATION IS NOT

One of the salient deductions that were determined in the systematic review is the asymmetrical developmental arc inherent to Bloom-aligned tasks:

Bloom-level classification is a discrete, closed-set classification task, which has constant labels, and thus is especially susceptible to discriminative learning paradigms. When trained on large, carefully annotated corpora, encoder-based architectures such BERT and RoBERTa additively improve their performance to be nearly human-reliable. [19], [38]. With an appropriate domain coverage and labeling balance, the problem of ambiguity in one cognitive level relative to another is effectively alleviated.

Generative tasks on the other hand are an open-ended problem that requires the model not just to identify but also decide on the level of cognitive challenge that should be engaged. Empirical studies repeatedly have found that generative models have the tendency of collapsing to safer, lower-order outputs, thus producing assessments, which, although seeming coherent, do not possess discriminative depth - a result that has been formalised as so-called ease bias. [35]. Even reasoning-based models still close this gap only when restricted to areas that are under strong logical constraints such as medicine, engineering, but they still face problems in interpretive or creative settings. [44], [46].

The current results show that, at this point, no single model structure can be able to offer credible classification and autonomous higher-order generation without some special constraints or human intervention.

C. PROBABILISTIC VOLATILITY AND THE LIMITS OF AUTONOMOUS ASSESSMENT

The main concern that has arisen in several studies is the concept of the so-called probabilistic volatility: the propensity to give different results on the same input when a LLM is run several times. Although this kind of variability can be acceptable in the context of exploratory learning tools, it is very dangerous in the context of assessment where consistency is a condition of fairness and validity.

High-stakes standardized tests such PISA, professional licensing exams indicate that the volatility rate rises rapidly when the cognitive level is higher, and the error margin is near-zero accuracy on the same material to outright inability to solve it [62]. The basic inefficiency of such instability is that generalist LLMs cannot be relied upon to assess higher-order thinking.

Pedagogically, this result supports the difference between the formative support and the summative judgment. Although it is possible that LLMs are useful in generating ideas, scaffolding, or exploring information, the evidence does not show that they can be used unsupervised in summative evaluation of higher-order learning outcomes.

D. PEDAGOGICAL RISK: COGNITIVE OFFLOADING AND THE EXPERTISE PARADOX

Beyond the technical constraints, there are significant pedagogical concerns brought up by the reviewed literature. First among them is cognitive offloading: a tendency of students to outsource intellectual work to AI systems, especially when it was seen as challenging or time-consuming [27]. By placing

the output of AI as authoritative and not provisional, there is a possibility of learners bypassing the metacognitive processes involved in the development of higher-order skills.

It is this phenomenon that is the basis of what Sorin et al [51] calls the Expertise paradox: models are most useful where learning needs are the least, and not where learning needs are the greatest due to the inconvenience they create. This issue is supported by empirical studies in the classroom, which report similar, or better performance on recall tasks but reduced improvement, or even negative improvement, on learning-intensive tasks using AI in the absence of pedagogical support [20], [87].

The findings of this work suggest that the educational usefulness of LLMs is not necessarily inherent to the technology, but it relies on the way tasks are designed around it. Structures like HOP and FPGA prove that, when AI is placed as a subject of critical questioning, instead of a solution provider, there are chances of maintaining a higher-order engagement. [27], [28].

E. METHODOLOGICAL IMPLICATIONS FOR EDUCATIONAL AI RESEARCH

The analysis also draws boundaries to methodological weaknesses of modern assessment procedures. Many of the studies use accuracy based measures that do not give sufficient consideration to the depth of thought, relevance of situation and the epistemic faithfulness, particularly at higher levels of Bloomian taxonomy. As a result, a vulnerability to the problem of false confidence, of having models interpreted as effective due to shallow task construction or slack benchmarking, still exists. [35].

Also, the emergence of using AI to validate the results produced by AI creates the issue of circular reasoning, which complicates the obscuration of real performance boundaries further on top of that same issue of performance obscuration [12]. Recent suggestions of extended taxonomies of critical thinking and multidimensional schemes of evaluation are hopeful paths, and they are not fully utilized in empirical studies [88]. In future research, the evaluation design will be approached as a research issue, but not as a supplementary reporting procedure.

F. TOWARD A DIVISION OF COGNITIVE LABOR

Taken together, the findings support a **division-of-labor model** for educational AI integration. Rather than seeking a universal model capable of performing all pedagogical functions, evidence favors specialized roles:

- **Discriminative models** such as fine-tuned BERT/RoBERTa for classification, auditing, and curriculum verification;
- **Reasoning-oriented generative models** for structured problem-solving in logic-constrained domains;
- **Efficient generative models** for drafting, personalization, and formative support;
- **Human educators** for judgment, synthesis, ethical oversight, and the cultivation of higher-order thinking.

This reframing transforms the question into one in which it can be seen that AI will replace educational judgment or how can AI redistribute cognitive labor and retain the intrinsic educational purpose?

G. SYNTHESIS

Altogether, the discussion makes it clear that the performance of LLMs is not always unsuccessful or successful with Bloom's Taxonomy; it is simply, task-, domain-, and architecture-dependent. The ability of classification is a full-fledged ability when the methodological conditions are correct. Generation is limited by the architectural and cognitive constraints especially at high levels. The issue of pedagogical risks is not existing in the use of AI, but rather in uncoordinated use that does not make technology relevant to the educational objectives.

These understandings give the conceptual gap between the empirical findings made above and the practical recommendations made in the conclusion.

V. CONCLUSION AND FUTURE DIRECTIONS

This SLR summarizes the evidence on 76 papers released in 2018-2025 to investigate the correspondence of LLMs with Bloom in the educational settings.

The results of the study provide an apparent generational and methodological gap in competence and trustworthiness. Generalist generative models such as GPT-3.5, GPT-4 are always effective at solving lower-order cognitive tasks such as recollection, understanding, routine application) but less reliable at solving higher-order classification and generation problems because of the volatility due to probabilities, semantic ambiguity, and limited pedagogical sensitivity. [8], [19].

Notably, the review has shown that the decade-old problem of Bloom-level classification has been overcome in a significant way, given clear methodological conditions. In the case of evaluating educational content with highly fine-tuned discriminative architectures, especially with BERT- and RoBERTa-based models trained on large domain-specific datasets, near-human accuracy (91%-98% accuracy) is always obtained. [19], [38], [39]. This creates Bloom-based classification as a mature and deployable ability of institutional functions including curriculum auditing, learning outcome verification, as well as accreditation processes.

Conversely, already, the pedagogically sound higher-order content generation is still skewed. Generative models perform well on the levels of Remember, Understand and Apply but lower on the levels of Analyze, Evaluate and Create, which substantiates a consistent Higher-Order Gap. [41]. Although reasoning oriented architectures (such as GPT-o1, Deepseek-R1) are reducing the performance gap between logic constrained domains like medicine and engineering [44], [45], there are major drawbacks in situations that need subjectivity, creativity, or interpretive reasoning [46].

IMPLICATIONS FOR EDUCATIONAL PRACTICE

All the evidence, however, supports a paradigm shift towards a pedagogy-first deployment paradigm. Instead of placing LLMs as decision-makers or instructional authorities, successful applications view the use of AI as a scaffold partner that facilitates the learning process without taking away the use of human judgment and agency. Structured integration systems, (HOP) and FPGA (Flipped Pedagogy and GenAI) model can indicate how AI can be used to support personalisation and efficiency without replacing the higher-order cognitive work that educational goals are focused on. [27], [28]

FUTURE RESEARCH DIRECTIONS

Based on the synthesised evidence, the following directions should be the priority of the future research:

- 1) **Expansion of Domain-Specific Training Corpora:** There is an need to develop large, publicly available datasets, comparable to the Monash CLO corpus [19] of the fine-tuning to STEM-dominated environments, which will cover business, social sciences, humanities, and interdisciplinary topics.
- 2) **Advanced Evaluation Frameworks:** The adoption of alternative multi-dimensional assessment systems, which describe the reasoning depth, situational suitability, and professional judgment, is a change that requires more than just accuracy-based measures and is consistent across the propositions outlined in broad critical-thinking taxonomies. [88].
- 3) **Hybrid Human-AI Architectures:** Design coordinated systems that integrate discriminative classifiers to perform auditing work, reasoning-capable models to generate a response to a complex problem solving situation, and efficient generative models to generate a routine draft, which all cannot be evaded by a structured human control.
- 4) **Longitudinal Pedagogical Studies:** Conduct a longitudinal research to determine the effect of AI-based learning conditions on the long-term acquisition of high-order cognitive capabilities.
- 5) **Stability and Reliability Mechanisms:** Probabilistic volatility, which is addressed through ensemble inference, retrieval-augmented generation and controlled sampling strategies are also a precondition to the safe implementation of educational systems.
- 6) **Cross-Domain Transfer Learning:** Test the claim that models that have been fine-tuned in one particular professional area can be successfully applied to other related areas without incurring extensive retraining.

Ultimately, practical evidences shows that the future of AI in learning does not lie with the replacement of human knowledge; rather, it represents a form of synergistic co-evolution with AI increasing the ability to scale and personalize learning, and an educator remaining responsible to encouraging deep, reflective and creative thinking.

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